

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: Artificial Intelligence

Code No: **CSC266** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **ntal**. There is a movie which is comedy. 3. How is informed search different from uninformed search? Create a state space with appropriate heuristics, now illustrate how hill climbing search expands nodes to reach a goal. Modify the state space heuristics and demonstrate when the hill climbing will not be complete. Section B Attempt any Eight questions [8×5=40] 4. What is intelligence? Describe the foundations of AI. 5. What is a rational agent? Justify with example how goal-based agent works? 6. How uniform cost search is used to search goal in the state space? Illustrate with example. 7. How can you represent knowledge using scripts? Create a knowledge base using script based on your own assumption. 8. What is reinforcement learning? Configure an ANN neuron to simulate OR gate. 9. What is robotics? How machine vision is used in robotics? [3+2] 10. Define fuzzy logic? Construct a fuzzy rule base expert system with your own considerations of fuzzy sets. 11. How is the minimax algorithm used in game search? Consider state space is defined by a collection of pairs like (A, B) representing paths between states A and B. Construct state space for following and use a minimax algorithm. (A, B), (A, C), (B, D), (D, E), (C, F), (C, G), (D, H), (D, I), (E, J), (F, K), (F, L), (G, M), (G, N). The utilities for states H, I, J, K, L, M, N are 1, 3, 2, 6, 3, 4, 1 respectively. 12. Justify which types of environments resembles following agents. a. Mission Game with fixed 5 states having two players. b. Tesla Driverless Robovan where road conditions are changing. c. Game Result Predicting Agent where current prediction state is independent of previous state.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. How can you relate synapse, dendrite, and axon in biological neural networks with the elements of artificial neural networks? Create a multi-layer ANN with input layer, hidden layer and output layer. Assume necessary inputs and weights to the ANN and illustrate a single iteration of back propagation algorithm to train the ANN.
2. What is Skolem constant? How is Skolemization done during resolution? Represent the following statements into FOPL. [2+3+5] All movies are not hit. Sarangi is a movie. All movies which has good script are hit. Sarangi has a good script but Sarangi is sentimental. There is a movie which is comedy.
3. How is informed search different from uninformed search? Create a state space with appropriate heuristics, now illustrate how hill climbing search expands nodes to reach a goal. Modify the state space heuristics and demonstrate when the hill climbing will not be complete

Section B

Attempt any Eight questions [8×5=40]

4. What is intelligence? Describe the foundations of AI.
5. What is a rational agent? Justify with example how goal-based agent works?
6. How uniform cost search is used to search goal in the state space? Illustrate with example.

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7. How can you represent knowledge using scripts? Create a knowledge base using script based on your own assumption.
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